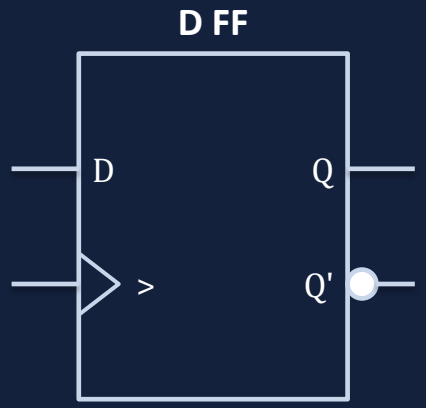
ENES 246 · DIGITAL LOGIC DESIGN

Lecture 19

Flip-Flops and Sequential Analysis



Edge-triggered D flip-flops · JK and T flip-flops · setup, hold and clock speed · analysis: state equations, tables and diagrams

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 5, §5.4–5.5

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Today’s roadmap

1 §5.4

D flip-flop

Edge-triggered storage

2 §5.4

JK and T

Characteristic equations

3 §5.4

Timing

Setup, hold, max frequency

4 §5.5

Analysis

State tables and diagrams

By the end of class you can…

- Contrast a latch and an edge-triggered flip-flop
- Use the characteristic equations of D, JK and T flip-flops
- Compute the maximum clock frequency and check hold time
- Derive state equations and a state table from a circuit
- Draw Mealy and Moore state diagrams

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Agenda for today's session



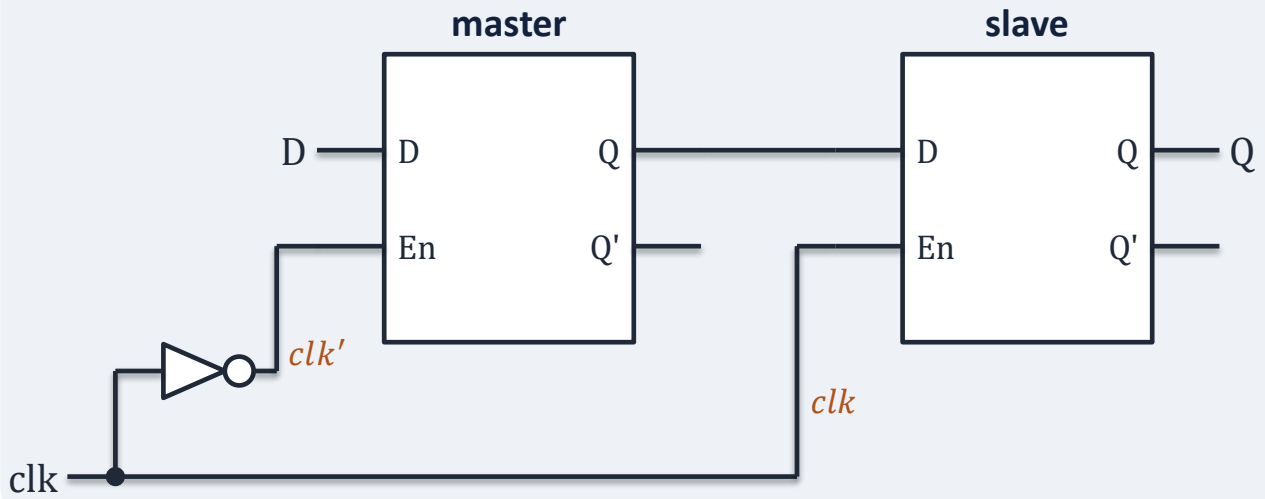
Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome; Homework 8 collected	—
0:05 – 0:25	Edge-triggered D flip-flop	Example 1
0:25 – 0:45	JK and T flip-flops	Example 2
0:45 – 0:55	Break	
0:55 – 1:10	Timing to scale; setup, hold and clock frequency	Example 3
1:10 – 1:40	Analysis of clocked circuits	Examples 4, 5 · Your turn
1:40 – 1:50	Conversions; concept check; wrap-up	—

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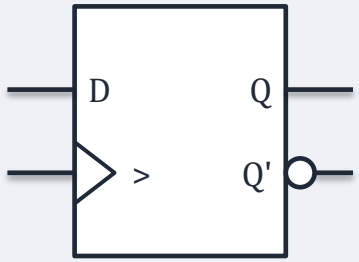
The edge-triggered D flip-flop

§5.4

Master-slave construction



Symbol (positive edge)



How it works

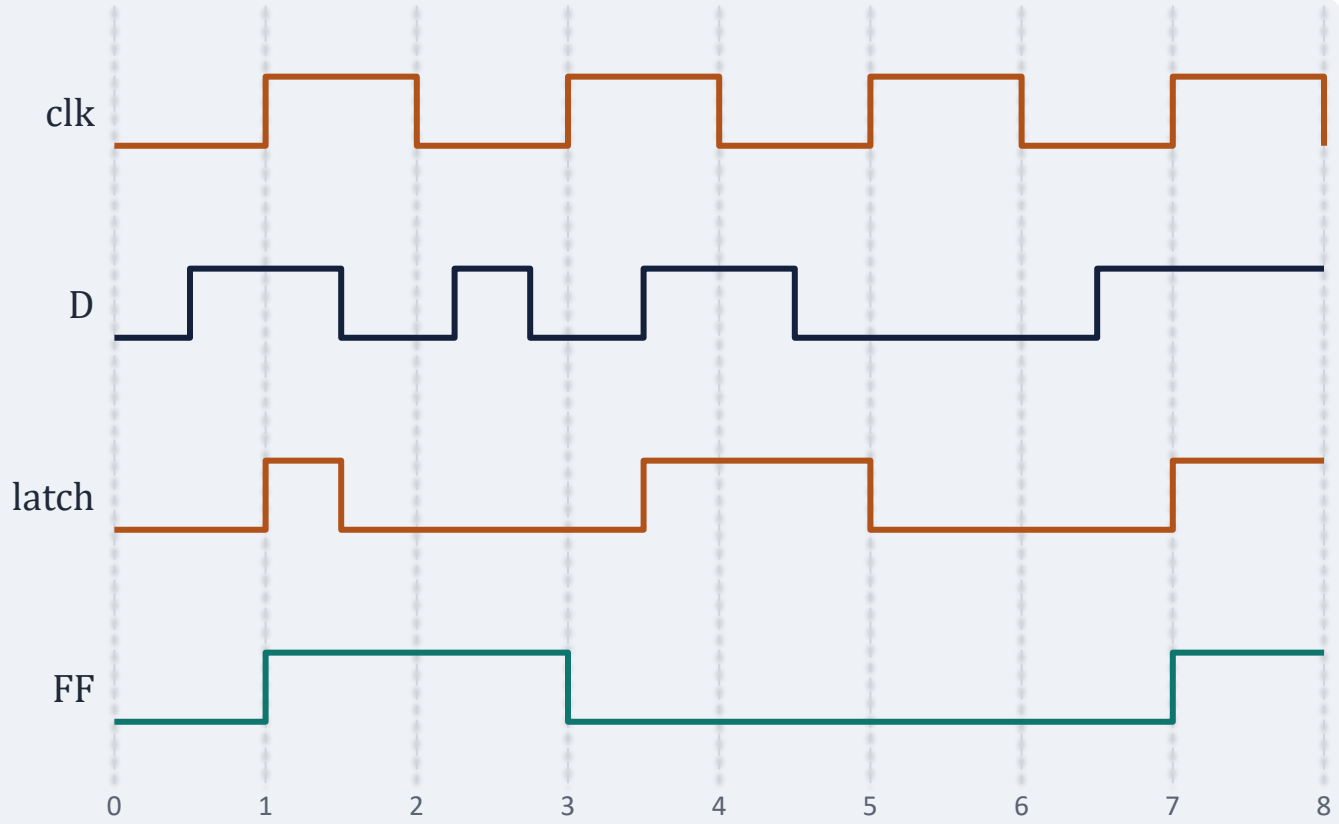
clk = 0: the master follows D, the slave holds. clk rises: the master freezes and the slave copies it. Q changes only at the rising edge: $Q(t+1) = D$.

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Example 1 — Latch versus flip-flop

§5.4

EXAMPLE The same D drives a D latch (En = clk) and a positive-edge D flip-flop. Both start at 0.



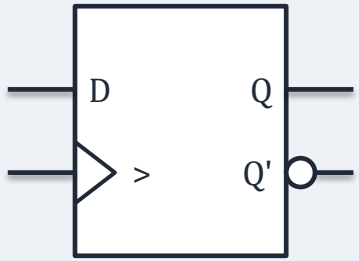
1. Latch: $Q = D$ while clk = 1; holds while clk = 0
2. 1–2: copies D: 1, then 0 at 1.5
3. 3–4: $D = 0$, then 1 at 3.5 → $Q = 1$
4. 5–6: $D = 0 \rightarrow Q = 0$; 7–8: $D = 1 \rightarrow Q = 1$
5. FF: samples D only at rising edges 1, 3, 5, 7
6. Samples 1, 0, 0, 1 → Q = 1 at 1, 0 at 3, 1 at 7
7. The 2.25 pulse is missed by both (clk low)
8. One sample per clock: no transparency

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JK and T flip-flops

§5.4

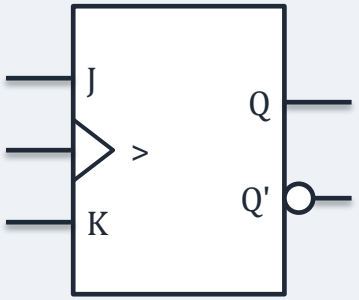
D flip-flop



D	Q(t+1)
0	0 reset
1	1 set

$$Q(t+1) = D$$

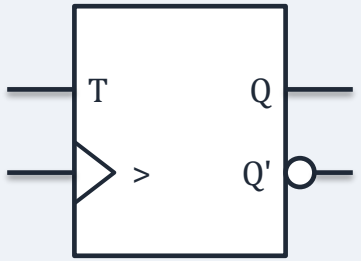
JK flip-flop



J K	Q(t+1)
0 0	Q hold
0 1	0 reset
1 0	1 set
1 1	Q′ toggle

$$Q(t+1) = JQ' + K'Q$$

T flip-flop



T	Q(t+1)
0	Q hold
1	Q′ toggle

$$Q(t+1) = T \oplus Q$$

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Example 2 — Trace a JK flip-flop

§5.4

EXAMPLE

A JK flip-flop starts at $Q = 0$. Apply JK = 10, 11, 00, 01, 11 on five clock edges. Then build T and D from a JK.

Edge	J	K	Q after	Action
1	1	0	1	set
2	1	1	0	toggle
3	0	0	0	hold
4	0	1	0	reset
5	1	1	1	toggle

1. Rules: 00 hold, 01 reset, 10 set, 11 toggle
2. Edge 1: $JK = 10 \rightarrow$ set: $Q = 1$
3. Edge 2: 11 → toggle: $Q = 0$
4. Edge 3: 00 → hold: $Q = 0$; edge 4: 01 → reset: $Q = 0$
5. Edge 5: 11 → toggle: $Q = 1$
6. T from JK: tie $J = K = T \rightarrow TQ' + T'Q = T \oplus Q$
7. D from JK: $J = D, K = D' \rightarrow DQ' + DQ = D$
8. T from D: $D = T \oplus Q$ (one XOR)

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Break

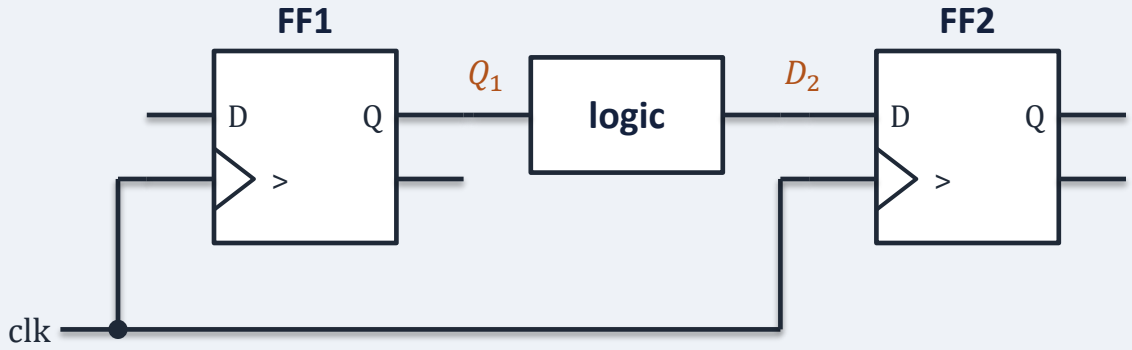
10 minutes — we resume at the 0:55 mark

Next: timing constraints and sequential analysis

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Flip-flop timing to scale

§5.4



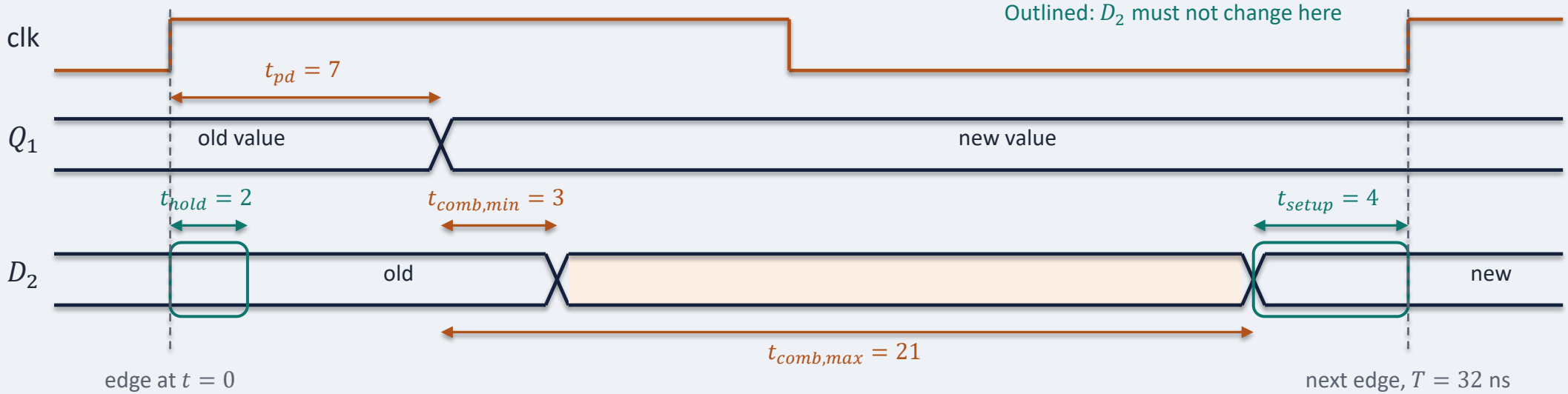
t_{pd}, propagation delay (clock-to-Q): from the clock edge until Q shows its new value

t_{comb}: delay through the logic (min = fastest path, max = slowest)

t_{setup}: D must be stable this long BEFORE the edge

t_{hold}: D must stay stable this long AFTER the edge

Here: setup margin $28 - 28 = 0$ ns; hold margin $10 - 2 = 8$ ns



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Example 3 — Setup, hold and speed

§5.4

EXAMPLE

Flip-flops: $t_{pd} = 7$, $t_{setup} = 4$, $t_{hold} = 2$ ns. Logic between them: longest 21 ns, shortest 3 ns.

Definitions

t_{pd}, propagation delay: from the clock edge until Q shows its new value (clock-to-Q).

t_{comb}: delay through the logic between the flip-flops.

Setup: D stable this long BEFORE the edge.

Hold: D stable this long AFTER the edge.

$$T \geq t_{pd} + t_{comb,max} + t_{setup}$$

$$t_{pd} + t_{comb,min} \geq t_{hold}$$

1. Setup: the slowest path must settle t_{setup} before the edge
2. $T \geq t_{pd} + t_{comb,max} + t_{setup} = 7 + 21 + 4 = 32$ ns
3. $f_{max} = 1/T = 1/32$ ns = **31.25 MHz**
4. Hold: the fastest change must come after t_{hold}
5. Earliest change at D: $t_{pd} + t_{comb,min} = 7 + 3 = 10$ ns
6. $10 \geq 2$ ✓; margin $10 - 2 =$ **8 ns**
7. Clock skew eats into both margins

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Analysis of clocked circuits

§5.5

1. Read the flip-flop input equations and output equations off the diagram.
2. Substitute into the characteristic equation(s) to get state equations.
3. List every present state and input combination: 2^{n+m} rows.
4. Fill in next state and output: the state table.
5. Draw the state diagram: circles for states, arcs for transitions.
6. Classify: Mealy (output depends on input) or Moore (state only).

Next-state rules

D: $Q(t+1) = D$

JK: $Q(t+1) = JQ' + K'Q$

T: $Q(t+1) = T \oplus Q$

Labels

Mealy arcs: input/output, e.g. 1/0.

Moore circles: state/output, e.g. 11/1.

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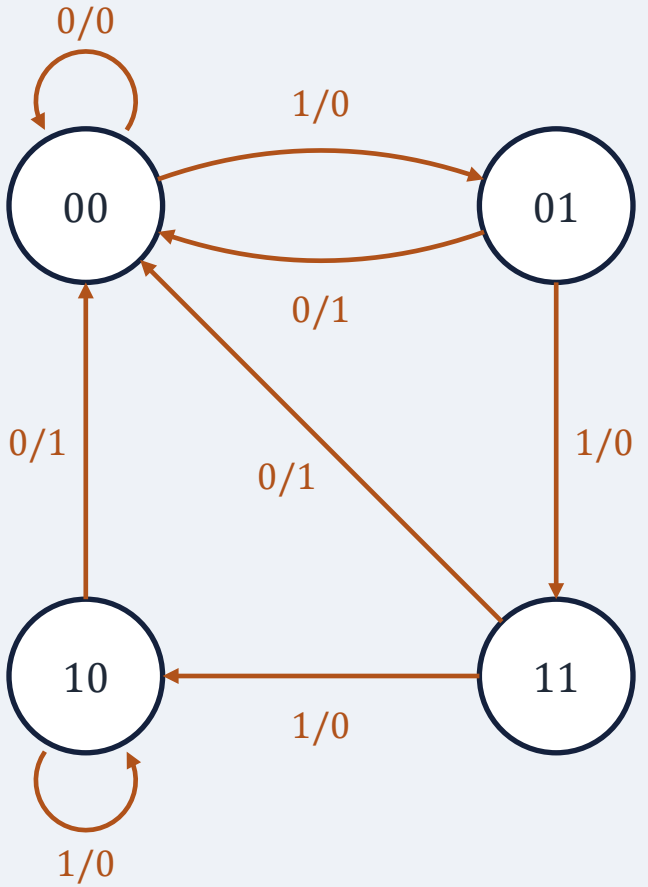
Example 4 — Analysis with D flip-flops

§5.5

EXAMPLE $D_A = Ax + Bx$, $D_B = A'x$, $y = (A + B)x'$. Find the state table and state diagram.

A B	x	A⁺ B⁺	y
0 0	0	0 0	0
0 0	1	0 1	0
0 1	0	0 0	1
0 1	1	1 1	0
1 0	0	0 0	1
1 0	1	1 0	0
1 1	0	0 0	1
1 1	1	1 0	0

1. State equations: $A^+ = Ax + Bx$, $B^+ = A'x$
2. Output: $y = (A + B)x'$
3. Each row: put A, B, x into all three
4. e.g. 01, $x = 1$: $A^+ = 0 + 1 = 1$, $B^+ = 1 \rightarrow 11$
5. Each row is one arrow, labelled x/y
6. Mealy: y depends on x



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Example 5 — JK flip-flop analysis

§5.5

EXAMPLE $J_A = B, K_A = Bx', J_B = x', K_B = A \oplus x$. Find the next states.

A B	x	JA KA	JB KB	A^+ B^+
0 0	0	0 0	1 0	0 1
0 0	1	0 0	0 1	0 0
0 1	0	1 1	1 0	1 1
0 1	1	1 0	0 1	1 0
1 0	0	0 0	1 1	1 1
1 0	1	0 0	0 0	1 0
1 1	0	1 1	1 1	0 0
1 1	1	1 0	0 0	1 1

1. Method 1: per row, evaluate J, K; apply the JK rule
2. Row 01, $x = 0$: $J_A = B = 1, K_A = Bx' = 1 \rightarrow$ toggle: $A = 1$
3. $J_B = x' = 1, K_B = A \oplus x = 0 \rightarrow$ set: $B = 1 \rightarrow$ next 11
4. Method 2: $Q^+ = JQ' + K'Q$ for each flip-flop
5. $A^+ = BA' + (Bx')'A = A'B + AB' + Ax$
6. $B^+ = x'B' + (A \oplus x)'B = B'x' + ABx + A'Bx'$
7. Both methods agree on all 8 rows ✓

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Your turn — T flip-flop circuit

§5.5

YOUR TURN

≈ 12 min

A circuit has T flip-flops A and B, input x and output y with $T_A = Bx$, $T_B = x$, $y = AB$.

(a) Write the state equations.

(b) Build the state table.

(c) Draw the state diagram. Is it Mealy or Moore? What does it do?

$Q(t+1) = T \oplus Q$

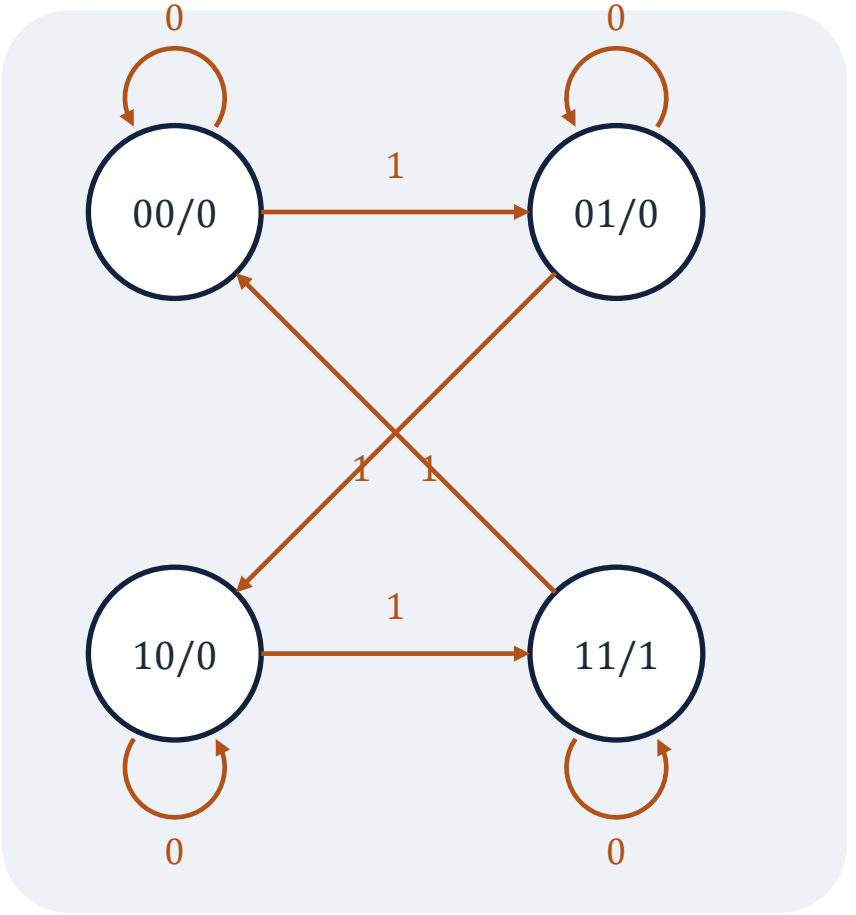
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Your turn — Solution

§5.5

SOLUTION

A B	x	A^+ B^+	y
0 0	0	0 0	0
0 0	1	0 1	0
0 1	0	0 1	0
0 1	1	1 0	0
1 0	0	1 0	0
1 0	1	1 1	0
1 1	0	1 1	1
1 1	1	0 0	1



1. T flip-flop: $Q^+ = T \oplus Q$
2. (a) $A^+ = A \oplus Bx$, $B^+ = B \oplus x$
3. (b) $x = 0$: both $T = 0$ → hold
4. $x = 1$: B toggles; A toggles when $B = 1$
5. So $x = 1$ counts 00 → 01 → 10 → 11 → 00
6. (c) $y = AB$ uses only the state: **Moore**
7. A 2-bit up counter; $y = 1$ in state 11

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Conversions and asynchronous inputs

§5.4

Building one flip-flop from another

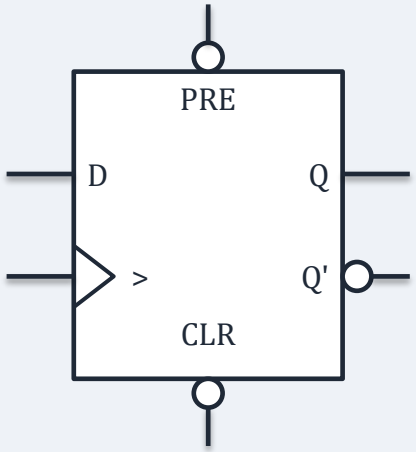
D from JK: $J = D, K = D'$

T from JK: $J = K = T$ (no gates)

T from D: $D = T \oplus Q$ (one XOR)

Method: set the needed next state equal to the available characteristic equation and solve for the inputs.

Preset and clear



Active-low PRE forces $Q = 1$, CLR forces $Q = 0$, at once, without the clock.

Used to start a system in a known state at power-up.

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Concept check

1. What does a flip-flop do that a latch does not?

2. What does a JK flip-flop do when J = K = 1?

3. Which timing constraint sets the maximum clock frequency?

4. How do you tell a Mealy machine from a Moore machine?

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Key takeaways

D

$$Q(t+1) = D$$

JK

$$Q(t+1) = JQ' + K'Q$$

T

$$Q(t+1) = T \oplus Q$$

Clock period

$$t_{pd} + t_{comb} + t_{setup}$$

Analysis

equations → table → diagram

Mealy / Moore

output uses input / state only

A flip-flop samples once per clock: that is synchronous design.

Homework #9 due at the start of Lecture 21

Next: state reduction and sequential design

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